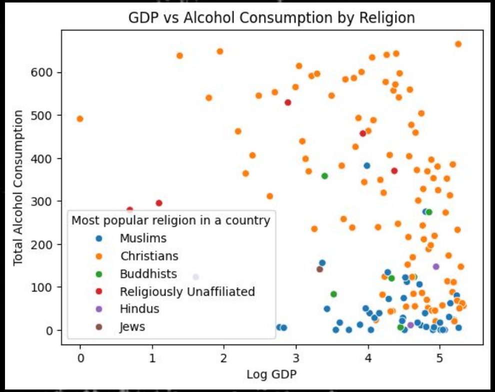
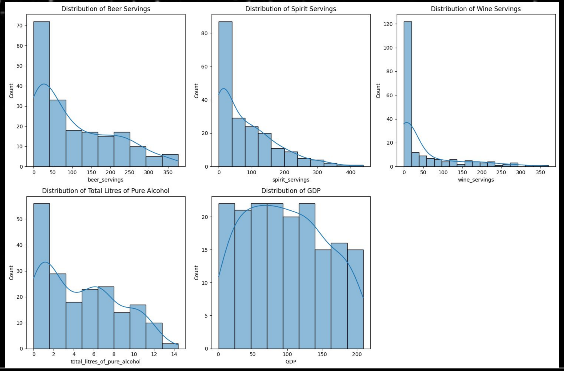
Is GDP an indicator of total alcohol consumption in a given country?



Null Hypothesis: There is no relationship between GDP and total alcohol consumption in a given country and religion does not play a role in that comparison.

Alternative Hypothesis: There is a relationship between GDP and alcohol consumption and the majority religion in a given country influences that affect.



Model:

Baseline Equation: $\text{Total Alcohol} = \beta0+\beta1(\text{logGDP})$

Modified Equation: Total Alcohol = $\beta0+\beta1(\text{logGDP})+\beta2(\text{Religion})+\beta3(\text{logGDPxReligion})+\epsilon$

Regression Results

Effect of GDP: p-value of 0.250

Effect of GDP on alcohol consumption per religion:

Buddhist: - 2.2252
Christian: -2.4678 (p-value 0.001)
Hindu: -0.9833
Judaism: -1.5780 (p-value 0.013)
Muslim: -0.9397

Conclusion:

Overall, we cannot reject the null hypothesis given our p-value.

However, majority religion does seem to have a stronger effect on total alcohol consumption in a county.